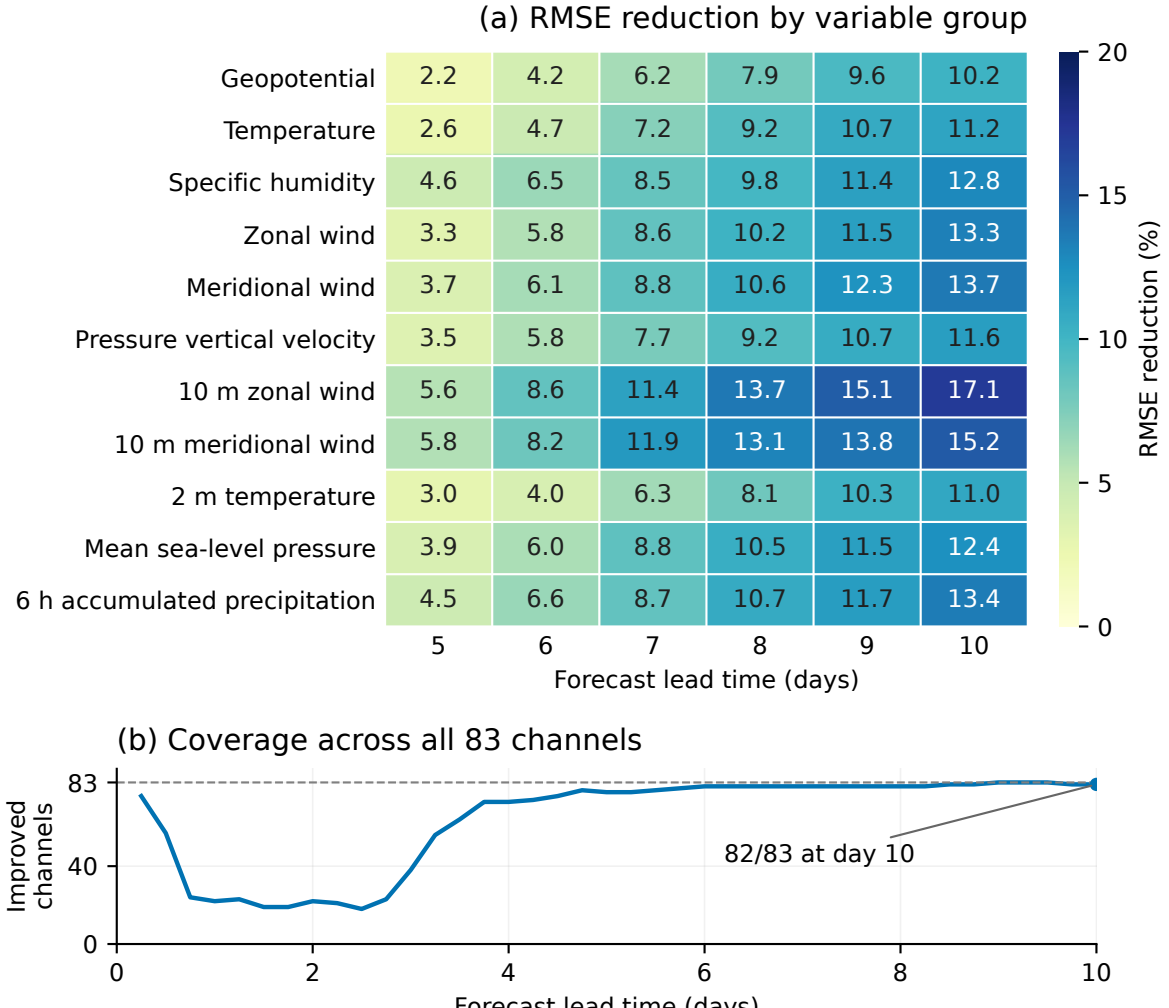
(a) RMSE reduction by variable group
Geopotential 2.2 4.2 6.2 7.9 9.6 10.2
Temperature 2.6 4.7 7.2 9.2 10.7 11.2
Specific humidity 4.6 6.5 8.5 9.8 11.4 12.8
Zonal wind 3.3 5.8 8.6 10.2 11.5 13.3
Meridional wind 3.7 6.1 8.8 10.6 12.3 13.7
Pressure vertical velocity 3.5 5.8 7.7 9.2 10.7 11.6
10 m zonal wind 5.6 8.6 11.4 13.7 15.1 17.1
10 m meridional wind 5.8 8.2 11.9 13.1 13.8 15.2
2 m temperature 3.0 4.0 6.3 8.1 10.3 11.0
Mean sea-level pressure 3.9 6.0 8.8 10.5 11.5 12.4
6 h accumulated precipitation 4.5 6.6 8.7 10.7 11.7 13.4
5 6 7 8 9 10
Forecast lead time (days)
RMSE reduction (%)
0 5 10 15 20
(b) Coverage across all 83 channels
Improved channels
83
40
0
82/83 at day 10
0 2 4 6 8 10
Forecast lead time (days)